Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 04)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Table 1 shows features of oil tanker design and operation that reduce the risk of oil pollution.

0 1 . 1 Complete Table 1 by explaining how each feature reduces oil pollution.

[5 marks]

Table 1

Feature	How it reduces oil pollution	
Double hull		
Inert gas system		
Separate oil and ballast tanks		
Twin engines / rudders		
Recirculation of oil during unloading (crude oil washing)		

0 1 . 2 Explain how each of the following reduces the risk of tanker accidents.

[2 marks]

Offshore shipping routes .....

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GPS / Automatic Identification System .....

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0 1 . 3 Describe how a **bund wall** prevents oil pollution from a storage tank.

[2 marks]

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0 1 . 4 Explain the purpose of an oil interceptor on the drainage system of a car park.

[1 mark]

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0 2 A factory discharges a persistent, water-soluble pollutant into a river. Table 1 shows the concentration of the pollutant measured at sites along the river on a day with normal river flow.

Table 1

Site	Distance downstream of discharge / km	Pollutant concentration / mg dm^{-3}	River flow / $m^3 s^{-1}$
A (upstream)	-2	0.0	12
B	0.5	4.8	12
C	5	3.9	15
D (below tributary)	12	1.6	36
E	30	1.4	38

0 2 . 1 Calculate the percentage decrease in pollutant concentration between site B and site E.

[2 marks]

Answer ☐ %

0 2 . 2 Explain the large decrease in concentration between sites C and D.

[2 marks]

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0 2 . 3 Explain why the concentration changes very little between sites D and E.

[2 marks]

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0 2 . 4 Suggest why the pollution might be more serious if the same discharge entered a lake instead of a river.

[2 marks]

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0 2 . 5 Explain why the factory might be permitted to discharge the effluent into the sea near a strong current but not into the river.

[2 marks]

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0 3 . 1 Explain **three** features of fossil fuels that made them ideal for the development of industrial society.

[3 marks]

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0 3 . 2 75 t of aviation fuel carries a Boeing 747 and its 400 passengers from London to New York. Calculate the mass of fuel used per passenger, in kg.

[2 marks]

Answer ..... kg

0 3 . 3 Explain why the total quantity of fossil fuels in existence is much greater than the quantity counted as **reserves**.

[3 marks]

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0 3 . 4 Cheap North Sea gas contributed to the closure of the UK's deep coal mines, although large coal deposits remained. Explain why those mines cannot now be reopened.

[2 marks]

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0 3 . 5 Explain **four** environmental impacts of the **extraction** of fossil fuels, as distinct from burning them.

[5 marks]

15

0 4 Table 1 shows typical noise levels of some sounds.

Table 1

Sound	Noise level / dB
Threshold of human hearing (at 1000 Hz)	0
Leaves rustling	20
Library	40
Normal conversation	60
Busy road	80
Chain saw	110
Jet engine at 30 m	140

0 4 . 1 Explain why a logarithmic scale is used to measure sound.

[2 marks]

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0 4 . 2 Use Table 1 to calculate how many times greater the sound intensity of a chain saw is than that of a busy road.

[2 marks]

Answer ☐ times

0 4 . 3 A machine produces 85 dB. A second identical machine is switched on next to it. State the new noise level and explain your answer.

[2 marks]

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0 4 . 4 Explain what is meant by the statement that 0 dB does not mean there is no sound.

[1 mark]

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0 4 . 5 State the frequency range of hearing for a typical young person and explain why the dB(A) scale is used for human hearing.

[3 marks]

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0 5 The Aral Sea in central Asia was once the fourth largest lake in the world. From the 1960s water from the two rivers feeding it (the Amu Darya and Syr Darya) was diverted to irrigate cotton and rice. Table 1 shows changes in the lake.

Table 1

Year	1960	1980	2000	2015
Lake area / km^2	68 000	52 000	24 000	7 000
Salinity / g dm^{-3}	10	18	45	over 100
Commercial fish catch / thousand tonnes	40	10	0	0

0 5 . 1 Calculate the percentage reduction in the area of the Aral Sea between 1960 and 2015.

[2 marks]

Answer ..... %

0 5 . 2 Explain why the salinity of the lake increased as the area fell.

[2 marks]

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0 5 . 3 Explain why the fish catch fell to zero, and describe **two** other environmental consequences of the shrinking of the lake.

[4 marks]

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0 5 . 4 Suggest why disagreements over the rivers continue between the countries of the region.

[2 marks]

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0 6 A student measured the mass of carbon stored per hectare in three land uses in the same region.

Table 1

Land use	Carbon in vegetation / t ha^{-1}	Carbon in soil dead organic matter / t ha^{-1}
Mature woodland	120	95
Permanent pasture	3	80
Arable (ploughed annually)	2	45

0 6 . 1 Calculate the total carbon lost per hectare when mature woodland is converted to arable farmland.

[1 mark]

Answer ☐ t/ha

0 6 . 2 Explain why the soil carbon in arable land is so much lower than in pasture, even though the vegetation carbon is similar.

[2 marks]

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0 6 . 3 Suggest **two** ways the arable farmer could increase soil carbon storage.

[2 marks]

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2 .....

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0 7 . 1 Describe **four** factors that should be considered when choosing a site for a dam and reservoir.

[4 marks]

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0 7 . 2 Give **one** advantage and **two** disadvantages of an estuary barrage compared with a reservoir on land.

[3 marks]

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Advantage .....

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Disadvantage 1 .....

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Disadvantage 2 .....

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0 7 . 3 Explain why desalination of seawater is only used in some countries.

[2 marks]

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0 7 . 4 Explain how artificial recharge can make aquifer supplies sustainable.

[1 mark]

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0 8 . 1 Which of the following is the main advantage of composting organic waste rather than landfilling it? Tick **one** box.

[1 mark]

A Aerobic decomposition produces little methane and returns nutrients to the soil ☒

B It produces more electricity ☐

C It removes heavy metals ☐

D It destroys asbestos fibres ☐

0 8 . 2 Explain why the ash from waste incinerators must be disposed of carefully.

[2 marks]

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0 8 . 3 Give **two** features of a well-managed landfill site that reduce pollution.

[2 marks]

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2 .....

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0 9 . 1 Explain what is meant by the **embodied energy** of a vehicle and how it can be reduced.

[3 marks]

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0 9 . 2 Describe **four** principles of designing vehicles for end of life.

[4 marks]

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0 9 . 3 Explain why there is an optimum speed for fuel efficiency in a car.

[2 marks]

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0 9 . 4 Give **one** way driver behaviour or vehicle systems can reduce fuel use in congested traffic.

[1 mark]

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1 0 . 1 Explain why the development of energy storage technologies is essential if renewable energy resources are to provide most of a country's electricity.

[9 marks]

Extra space

1 0 . 2 State what is meant by **peak shaving**.

[1 mark]

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Write an essay on one of the following topics.

11 . 1 Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country.

[25 marks]

OR

11 . 2 Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS